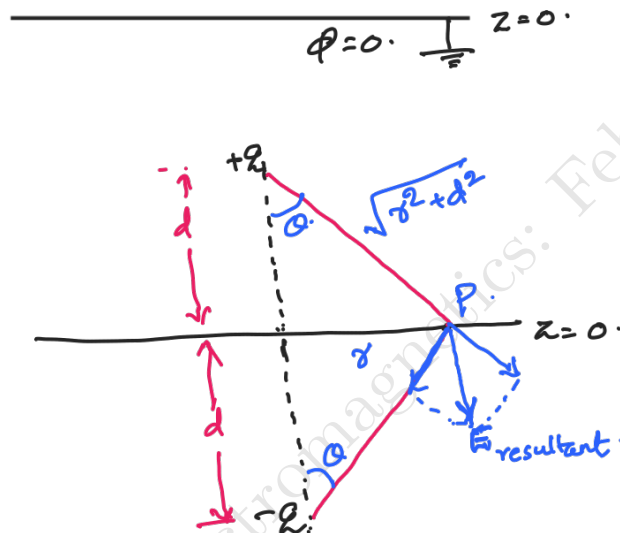


Method of images:. 2 $\phi ?$ Coulomb's law

$$\vec{E}(P) = \frac{Q}{4\pi\epsilon_0} \cdot \frac{1}{(r^2 + d^2)} \cdot \frac{(r\hat{r} - d\hat{z})}{(r^2 + d^2)^{3/2}}$$

unit vector in the direction of electric field

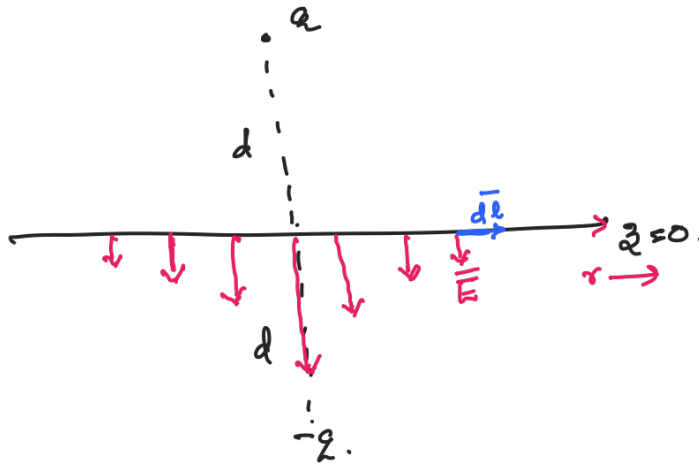
$$+ \frac{Q}{4\pi\epsilon_0} \cdot \frac{1}{(r^2 + d^2)} \cdot \frac{(-r\hat{r} - d\hat{z})}{(r^2 + d^2)^{3/2}}$$

-ve sign is neglected here as it is taken care by direction of $\vec{E}$.

$$\boxed{\vec{E}(P) = \frac{Q}{4\pi\epsilon_0} \cdot \frac{(-2d)\hat{z}}{(r^2 + d^2)^{3/2}}}$$

→ verify with our previous lecture results.





$\phi \rightarrow 0$ at far away. (By point charge & Coulomb's law definition.)

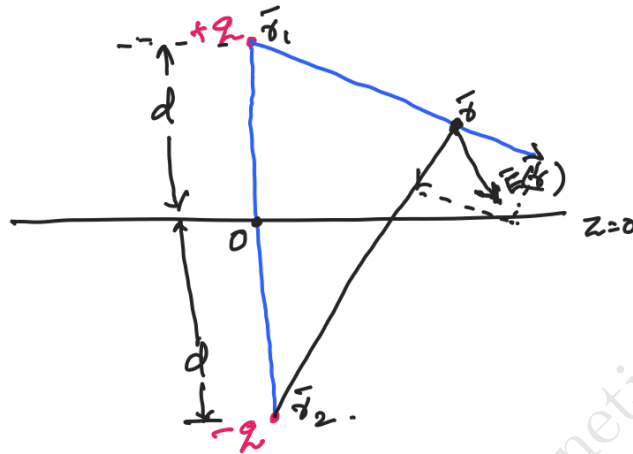
ϕ at $z=0$ plane (at any point in the plane)

$$\text{ie, } \phi(r, z=0) - \phi(\infty) = \int_0^\infty \vec{E} \cdot d\vec{l} \quad \vec{E} \perp d\vec{l}$$

$$\boxed{\phi(r, z=0) = 0} \quad \checkmark$$

Now by Uniqueness theorem we know that the solution at $z > 0$ is the same for both the problems (a) with grounded conductor & point charge & (b) with two point charges similar to dipole symmetrically situated above & below $z=0$ plane).

Now in the above problem, Electric field anywhere say $\vec{r}$.

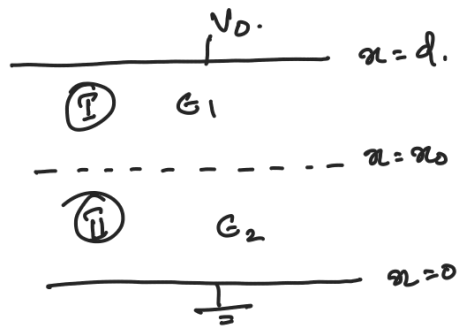


$$\begin{aligned}\vec{r}_1 &= (0, 0, d) \\ \vec{r}_2 &= (0, 0, -d) \\ \vec{r} &= (x, y, z) \\ |\vec{r} - \vec{r}_1| &= \sqrt{x^2 + y^2 + (z-d)^2}\end{aligned}$$

$$\begin{aligned}E(\vec{r}) &= \frac{q}{4\pi\epsilon_0} \left[\frac{\vec{r} - \vec{r}_1}{|\vec{r} - \vec{r}_1|^3} - \frac{\vec{r} - \vec{r}_2}{|\vec{r} - \vec{r}_2|^3} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{x\hat{i} + y\hat{j} + (z-d)\hat{k}}{(x^2 + y^2 + (z-d)^2)^{3/2}} - \frac{x\hat{i} + y\hat{j} + (z+d)\hat{k}}{(x^2 + y^2 + (z+d)^2)^{3/2}} \right]\end{aligned}$$

* Basically a complex problem is brought down to a situation where we can solve Coulomb's equation! by applying uniqueness theorem & method of images.

Uniqueness Theorem: Applied to solve capacitance problems.



$\vec{D}$ is continuous across all the boundary which are free of free charges.
 $\Rightarrow \vec{D}$ is constant.

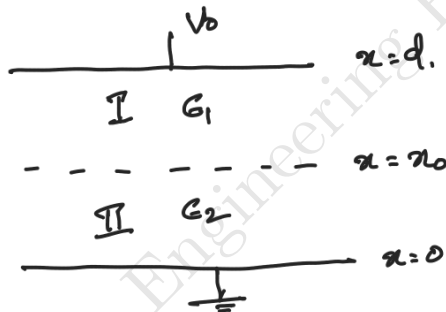
$$\Rightarrow E_x = \begin{cases} \frac{D_0}{\epsilon_1} & ; x_0 < x < d \\ D_0/\epsilon_2 & ; 0 < x < x_0 \end{cases}$$

Ref Lec' 11 §12.
Using Gauss law
($\vec{E} \& \vec{D} \Rightarrow$ Constant)

We solved for potential ϕ_0 ,

$$\phi = \begin{cases} \frac{D_0}{\epsilon_2} x_0 & ; 0 < x < x_0 \\ \frac{D_0}{\epsilon_2} x_0 + \frac{D_0}{\epsilon_1} (d - x_0) & ; x_0 < x < d \end{cases}$$

An alternate method:



$$E_x = -\frac{d\phi}{dx}$$

$$\nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} = 0 \quad (\text{charge free})$$

$$\Rightarrow \frac{\partial(\epsilon E_x)}{\partial x} = 0$$

$\Rightarrow$ Homogeneous, isotropic dielectric $\Rightarrow \epsilon \rightarrow$ constant

$$\Rightarrow \epsilon \frac{\partial(-\frac{\partial\phi}{\partial x})}{\partial x} = 0$$

$$\Rightarrow \boxed{\frac{\partial^2 \phi}{\partial x^2} = 0} \leftarrow \text{Solving Laplace's Eqn.}$$

$$\Rightarrow \frac{\partial^2 \phi}{\partial x^2} = 0 \rightarrow \text{1D Laplace's Eqn.}$$

$$\phi = C_1 x + C_2$$

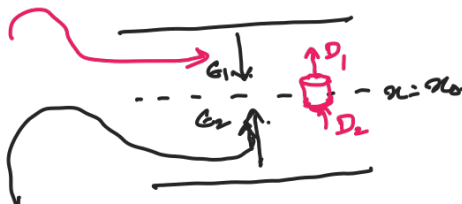
I region: $\phi = C_1 x + C_2$
 $\phi(d) = V_0$
 $\Rightarrow V_0 = C_1 d + C_2$
 $C_2 = (V_0 - C_1 d)$

II region: $\phi = C_3 x + C_4$
 $\phi(0) = 0 \Rightarrow$
 $C_4 = 0$
 $\Rightarrow \phi = C_3 x$

I: $\phi = C_1 x + (V_0 - C_1 d)$

II: $\phi = C_3 x$

Now if we know $\phi(x_0)$ we can know C_3 .

$$\phi = C_1 x_0 + (V_0 - C_1 d) = C_3 x_0$$


$$C_1 x_0 + (V_0 - C_1 d) = C_3 x_0 \quad \text{--- (1)}$$

$$\bar{D}_2 = \bar{D}_1 \quad \text{--- (2)} \quad \text{--- Gauss' theorem.}$$

$$D_2 = -\epsilon_2 \frac{\partial \phi}{\partial x} = -C_3 \epsilon_2 \quad (\text{In Region II})$$

$$D_1 = -\epsilon_1 \frac{\partial \phi}{\partial x} = -C_1 \epsilon_1 \quad (\text{In Region I})$$

$$D_2 = D_1 \Rightarrow C_3 \epsilon_2 = C_1 \epsilon_1$$

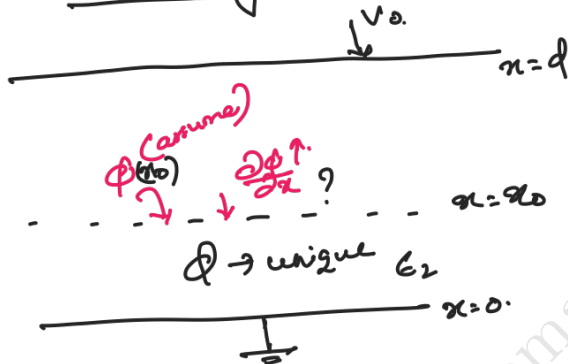
$$C_3 = \frac{\epsilon_1}{\epsilon_2} C_1$$

$$\Rightarrow C_1 x_0 + (V_0 - C_1 d) = \frac{\epsilon_1}{\epsilon_2} C_1 x_0$$

$$\Rightarrow C_1 \left(x_0 - d - \frac{G_1 x_0}{G_2} \right) = -V_0.$$

$$C_1 = \frac{V_0}{d - \left(1 - \epsilon_1/\epsilon_2\right) x_0}.$$

Where did we apply uniqueness theorems:



This iteration applying uniqueness theorems enables us to find the solution in entire region.

1. Assume ϕ at $x=x_0$.
 $\Rightarrow \phi$ unique $0 < x < x_0$.

2. Then $\frac{\partial \phi}{\partial n}$ is known.
 $\Rightarrow \phi$ is unique in $x_0 < x < d$.

3. ϕ in $x_0 < x < d$
 $\Rightarrow \phi$ at $x=x_0$ is known.

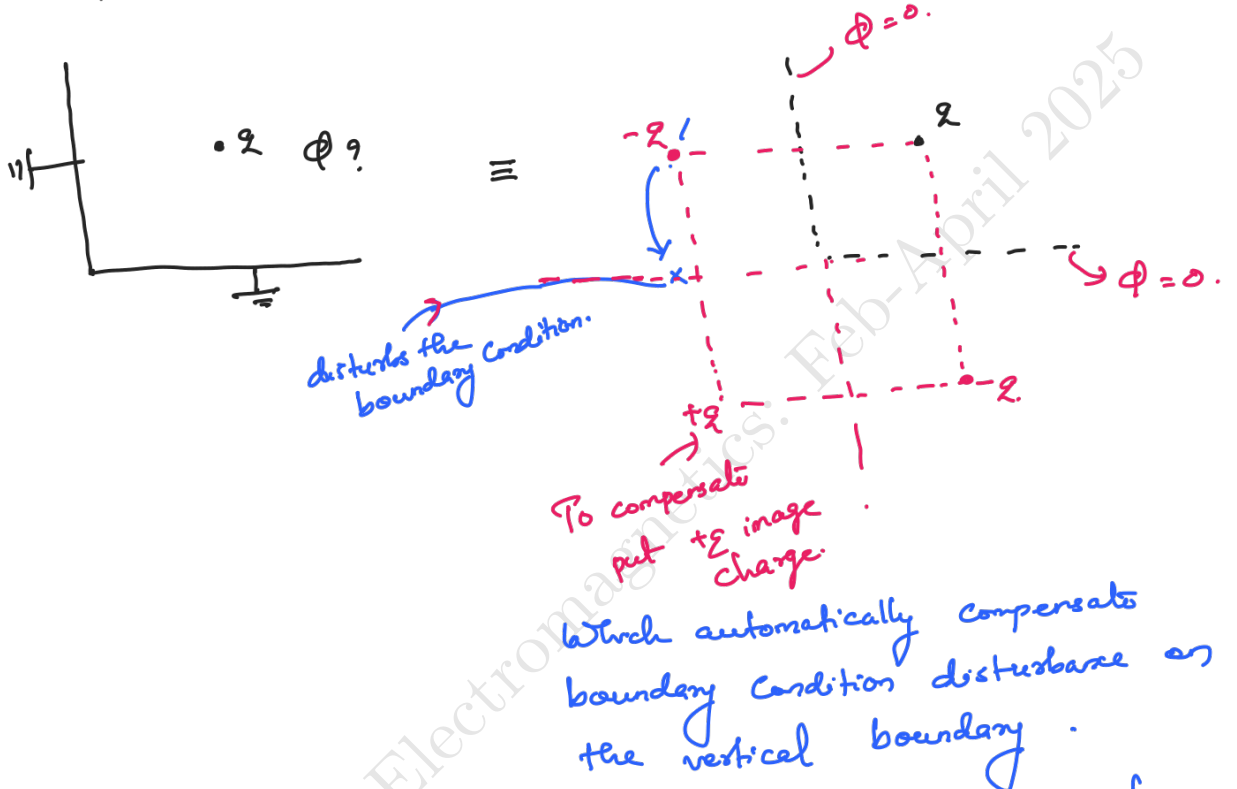
4. $\Rightarrow$ Again ϕ unique at $0 < x < x_0$.

Thus we know entire solution everywhere

Power of Uniqueness theorems!

Method of Images : more examples.

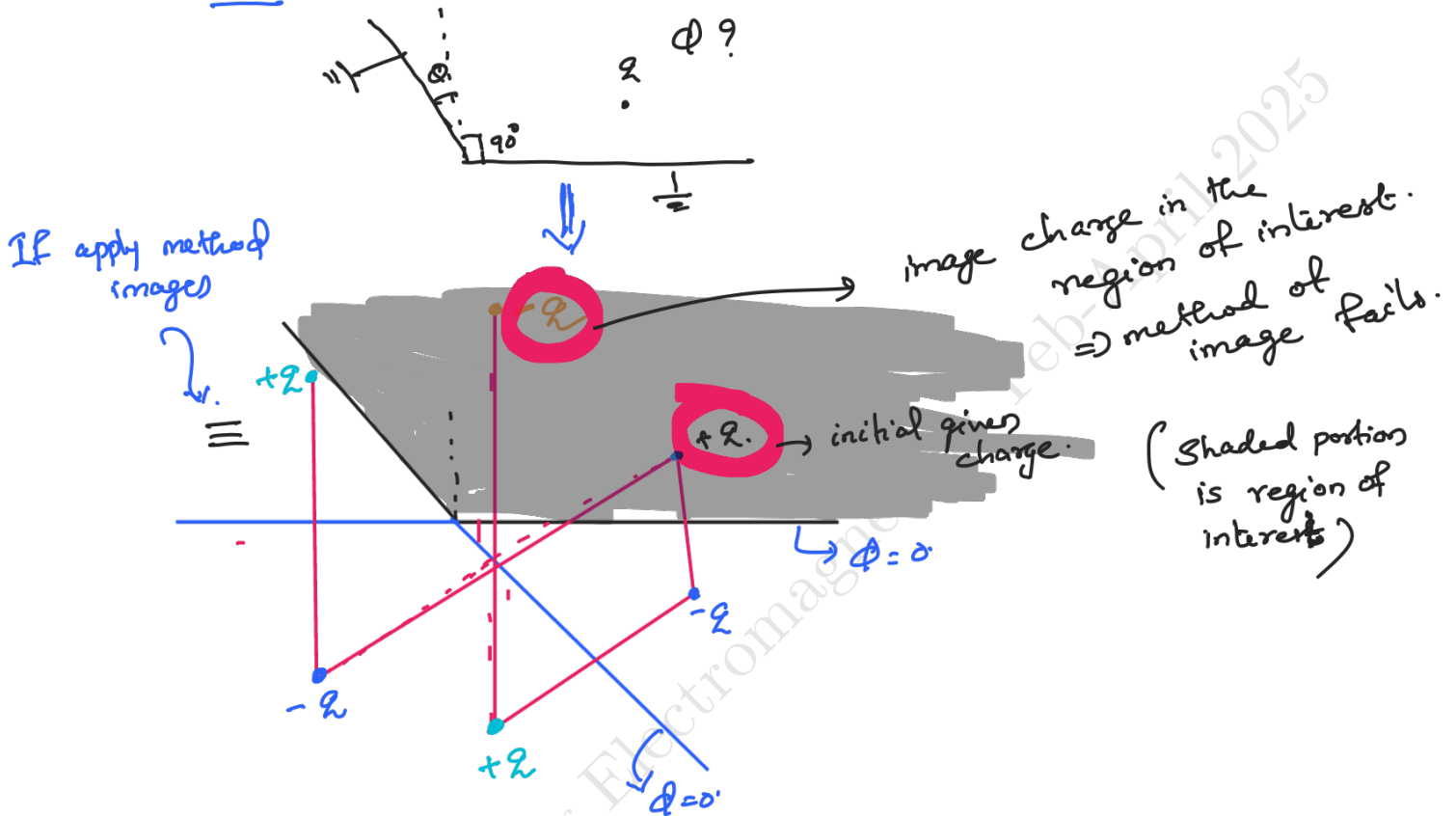
1.



Now the problem is brought down to 4 point-charge problem.

$$\Rightarrow E(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^4 \frac{q_i (\vec{r} - \vec{r}_i)}{|\vec{r} - \vec{r}_i|^3}$$

2. Problem 2:



Such problems of poisson's Eqn $\rightarrow$ solved using
Green's function (Topic for advanced course).

